

SaaS Attendance Management

Software System Documentation

SYSTEM DIAGRAMS & DATABASE SCHEMAS

Target Environment: Docker Containerization

Architecture: Django REST + React SPA

Date: July 2026

Chapter 1: Introduction & Project Scope

This software application is a secure, multi-tenant Software-as-a-Service (SaaS) Attendance Management Web Application. It enables educational institutions to automate and track student attendance using dynamic QR code verification. The application is built using a distributed containerized architecture consisting of a Python/Django backend, a React SPA frontend, and a MySQL database backend.

Key System Objectives

- Prevent Spoofing: QR codes rotate dynamically every 20 seconds, preventing students from marking attendance remotely or sharing screenshots.
- Tenant Scoping: Universities are segmented at the database level. Each administrator, teacher, and student can only access resources matching their authorized tenant.
- Complete Audit Controls: Administrators possess full CRUD capabilities over users, universities, courses, and active classes.

Chapter 2: System Architecture & Component Diagram

The system runs inside an isolated Docker Compose network containing three distinct components:

1. Web Client (SPA Browser): React single-page app containing user panels for scanning, directories, and dashboard metrics.
2. Nginx Reverse Proxy: Exposes port 80 to clients, routes static assets, and forwards API queries to the Django REST framework.
3. Django REST API: The Gunicorn server running Gunicorn/WSGI, resolving logins, sessions, and database mutations.
4. MySQL Database: Persists universities, user permissions, courses, departments, semesters, and sections.

Chapter 3: Database ERD (Entity-Relationship Diagram)

To accommodate academic routing, the database implements a strict four-layer tenant hierarchy: University (Institution) -> Department -> Semester -> Section.

Core Entities & Schema Relations

- Institution (University): Segmented tenant. All courses and user roles belong here.
- Department: Multiple departments (e.g. Computer Science, Software Engineering) exist under each University.
- Semester: Each department contains semesters (e.g. Semester 1 to Semester 8).
- Section: Semesters contain sections (e.g. Section A to Section Z).
- CustomUser: Has fields for email, password hash, role (admin/teacher/student), and foreign keys mapping them to their University, Department, Semester, and Section.
- Course: Connects to a teacher profile and the linked University.
- Enrollment: Links student user IDs to course IDs to tracks registration.
- AttendanceSession: Lives for a course and handles generating live QR tokens.
- AttendanceRecord: Stores student marking status ('present' or 'absent') and timestamps for a session.

Chapter 4: Core Features & Business Logic

1. Secure Live Attendance (Anti-Proxy System)

Allows teachers to take attendance instantly. The system generates a live QR code on the teacher's screen that automatically refreshes every 10 seconds. Because the code changes quickly, students cannot screenshot it to friends remotely.

2. Device Binding & Fraud Prevention

Ensures one student equals one phone. When a student first logs in, their account is permanently bound to their specific smartphone. If they switch devices legitimately, they must complete an email-verified 'Device Rebind' process.

3. Smart Attendance Alerts & Bulk Emails

Automatically categorizes students into three risk tiers based on their attendance percentage (Critical < 25%, Severe 25-50%, Warning 50-75%). Instead of sending individual emails, teachers can click 'Send Bulk Notice' to dispatch a single generic warning email with all at-risk students securely in the BCC field.

4. Admin Control & Excel Data Imports

Admins can upload Excel spreadsheets to bulk-create students and teachers. The system runs a 'Dry Run' to validate all emails and department mappings, highlighting errors before any data is permanently saved to the database.

5. Multi-Tier Hierarchy

Data is structured as a tree: Institution -> Department -> Semester -> Section. This ensures teachers only access their designated subset of students, keeping the software organized and secure.